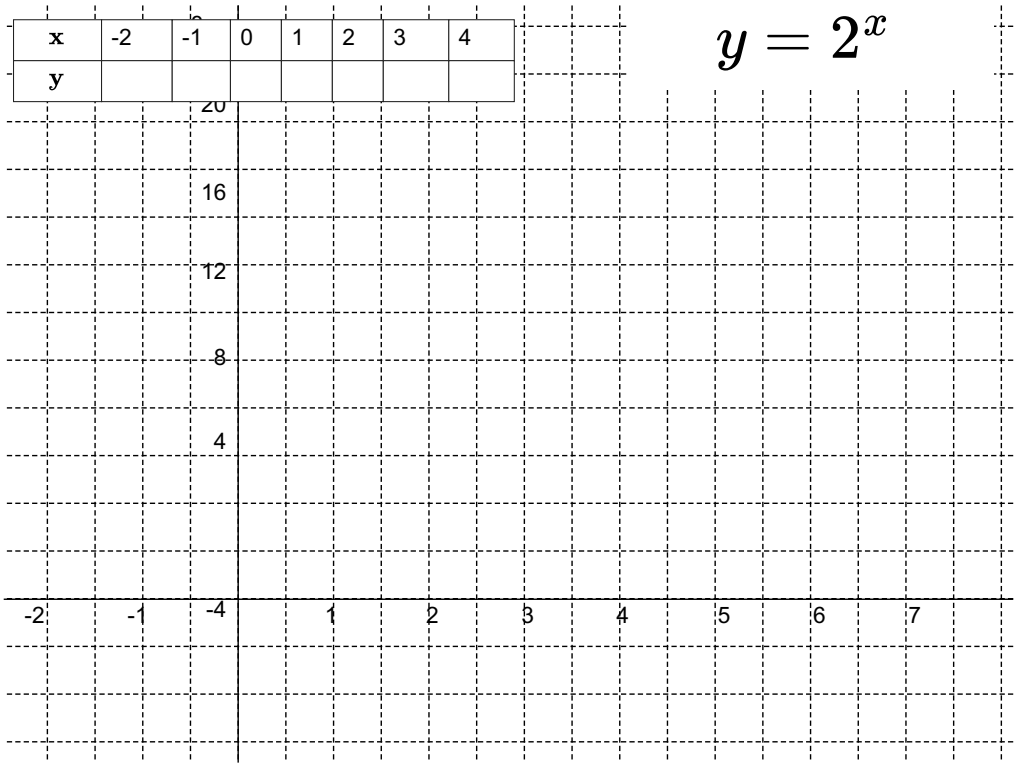
1.24 Graph Sketching: Exponentials and Logarithms

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x	-2	-1	0	1	2	3	4
y							

$y = 2^x$

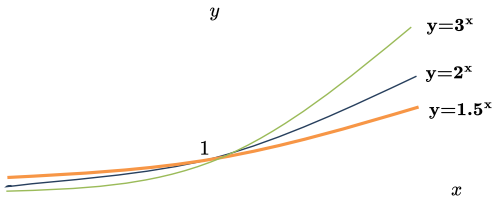


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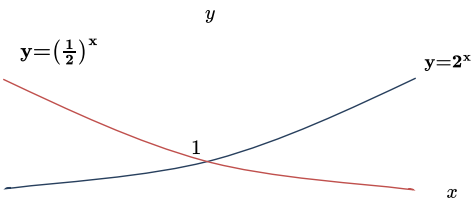
Contrasting exponential graphs

5

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$



On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$



Three important notes:

- $y = 2^x$ is said to be "**exponential growing**" whereas $y = \left(\frac{1}{2}\right)^x$ is said to be "**exponentially decaying**", because it's getting smaller (halving) each time x increases by 1.
- $y = \left(\frac{1}{2}\right)^x$ is a reflection of $y = 2^x$ in the line $x = 0$.
  Proof:
  If $f(x) = 2^x$, $f(-x) = 2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$
- $\left(\frac{1}{2}\right)^x$ would usually be written 2^{-x}.

You should therefore in general be able to recognise and sketch the graph $y = a^{-x}$.

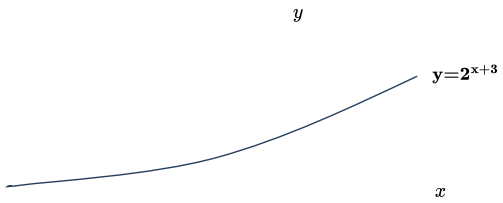
1.24 Graph Sketching: Exponentials and Logarithms

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Graph Transformations

6

Sketch $y = 2^{x+3}$



The 'change' to 2^x is 'inside the function' (i.e. input x is replaced with $x + 3$).

Ensure you work out the new y-intercept.

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Exercise 14A (Pages 313-314)

7

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

1.24 Graph Sketching: Exponentials and Logarithms

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x	0.25	0.5	1	2	4	8
y						

$y = \log_2 x$

